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Natural Language Processing

Session 3: Text Representation I
Vector-based approaches

GRAD-E1282 · Fall Semester 2026
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Session 1 in one slide: the shape of an NLP system

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LANGUAGE → REPRESENTATION → MODEL → EVALUATION

today

WHAT NLP DOES

- classify — a label or a score
- find / extract — text or facts
- generate — language from context

Real systems chain all three.

HOW IT IS BUILT

- rules & heuristics
- machine learning
- deep learning / transformers

Newer does not mean replacement.

WHAT LANGUAGE IS MADE OF

- sound / symbols
- word form
- structure (syntax)
- meaning in context

Each task reads a different level.

A bigger model cannot rescue a poorly specified problem.

Every representation makes some information visible, and hides the rest.

Session 2 in one slide: what counts as a unit

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RAW TEXT → PATTERNS → TOKENS → NORMALIZED TOKENS

REGEX

Describes a set of strings, not one string.

TOKENIZATION

A choice, not a fact. Rules, or units learned from corpus statistics.

NORMALIZATION

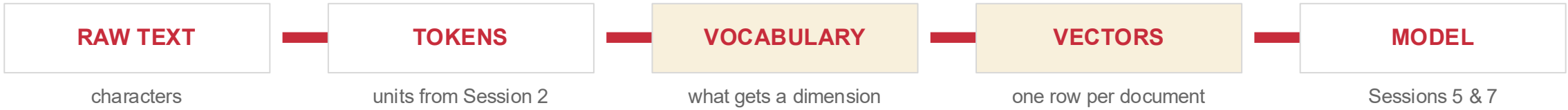
Deliberate information loss, useful or harmful, depending on the task.

corpus	document	token	vocabulary
the texts we analyse	one text within it	the unit of analysis	the distinct tokens kept

Today: count the units, weight the counts, and treat the result as a vector.

Today: from tokens to vectors

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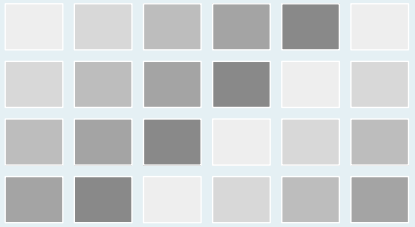
BY THE END OF TODAY

1. Explain why text representation is a modeling choice, not a file format.
2. Build and interpret one-hot, bag-of-words, n-gram and TF-IDF vectors.
3. Compare documents with cosine similarity — and say what each representation hides.

Why text representation is the hard part

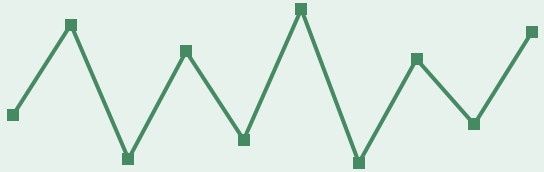
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IMAGE



pixel intensities → matrix

SPEECH



sample amplitudes → vector

TEXT

“Policy changed today.”

What should each number mean?

Text representation = choosing a numerical form that makes useful linguistic information available to the model.

The vector space model: vocabulary becomes dimensions

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TOY CORPUS

D1	dog bites man
D2	man bites dog
D3	dog eats meat
D4	man eats food

VOCABULARY V

dog	bites	man
eats	meat	food

|V| = 6 → six shared dimensions

D1 IN THAT SPACE

dog	bites	man	eats	meat	food
1	1	1	0	0	0

"dog bites man" → [1, 1, 1, 0, 0, 0]

CORPUS

The collection of documents under analysis. Here: D1–D4.

TOKEN

A discrete unit from tokenization: word, subword, character, punctuation.

VOCABULARY V

The set of distinct features kept from the corpus. Here |V| = 6.

Vector space model: every vocabulary feature becomes one shared dimension across all documents.

One-hot encoding: one word, one active dimension

	dog	bites	man	eats	meat	food
dog	1	0	0	0	0	0
bites	0	1	0	0	0	0
man	0	0	1	0	0	0

D1 = a sequence of three one-hot vectors

```
[[1 0 0 0 0 0]
 [0 1 0 0 0 0]
 [0 0 1 0 0 0]]
```

A 3-word document gives 3 vectors; a 10-word document gives 10.

WHY IT IS SELDOM USED TODAY

Sparse

|V| grows with the corpus; almost every entry is 0, and sparsity invites overfitting.

Not fixed length

Longer text → longer representation. Most learners need equal-length feature vectors.

Semantically blind

run, ran and apple are equally far apart. Words are atomic, unrelated units.

No OOV handling

An unseen word has no dimension. The only fix is to rebuild the vocabulary and retrain.

Intuitive and lossless at the token level, and almost unusable as document features.

Bag of Words: one fixed-length vector per document

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D1: "DOG BITES MAN"

dog	bites	man

COUNT INTO THE FIXED VOCABULARY ORDER

dog	bites	man	eats	meat	food
1	1	1	0	0	0

D1 → [1, 1, 1, 0, 0, 0]

WHAT WE GAIN

- fixed length for any document
- simple, fast, interpretable dimensions
- documents sharing words land closer together

WHAT WE GIVE UP

- word order and syntax
- similarity between different words
- still sparse, still no OOV handling

Connection to one-hot

BoW aggregates the token one-hot vectors of a document. Count BoW sums them, so "dog and dog are friends" gives dog = 2.
Binary BoW keeps only presence or absence — CountVectorizer(binary=True) — which research has found useful for sentiment tasks.

Despite its shortcomings, BoW remains a standard baseline , especially for text classification.

What exactly does Bag of Words forget?

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		dog	bites	man	eats	meat	food
D1	dog bites man	1	1	1	0	0	0
D2	man bites dog	1	1	1	0	0	0

same vector

“dog bites man” and “man bites dog” describe opposite events.
Bag of Words assigns them identical representations, because it never recorded order.

This is not a bug in the code. It is a property of the representation — and therefore a ceiling on any model built on top of it.

Bag of n-grams: putting a little order back

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D1 → BIGRAMS

`dog bites man`

`dog bites` | `bites man`

D2 → BIGRAMS

`man bites dog`

`man bites` | `bites dog`

D1 and D2 no longer collide.

THE COST: THE FEATURE SPACE GROWS

Bigram vocabulary of the whole toy corpus

`dog bites` · `bites man` · `man bites` · `bites dog` · `dog eats` · `eats meat` · `man eats` · `eats food`

8 features for 4 documents of 3 words — every document now uses just 2 of them.

n = 1 → **larger n**

less context	more local order	far more features	more sparsity

BoW is simply the n = 1 case. CountVectorizer(ngram_range=(1, 2)) mixes unigrams and bigrams.

n-grams buy back short, local order. They do not solve sparsity, and they still cannot handle unseen words.

Session Break

When we return: why raw counts are not enough → TF-IDF, cosine similarity and a tiny search engine.

TF-IDF: not all words are equally informative

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TOY EXAMPLE: TERM FREQUENCY IN TWO WIKIPEDIA SECTIONS ON ALGORITHMIC BIAS

	"bias"	"and"	
Intro section	0.0167	0.0292	more "and" than "bias"
Discrimination section	0.0022	0.0303	14× more "and" than "bias"

By raw term frequency, both sections are "about" the word and.

1 · TERM FREQUENCY (TF)

How prominent is this term inside this document?
Normalised by document length, so long documents do not win automatically.

more local occurrences → higher weight

2 · INVERSE DOCUMENT FREQUENCY (IDF)

How surprising is it that this term appears at all?
Terms common across the corpus are weighed down; rare ones are weighed up.

appears in fewer documents → higher weight

TF-IDF(t, d) = TF(t, d) × IDF(t) **frequent here, rare elsewhere → characteristic of this document**

TF × IDF: how the weight is built

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TERM FREQUENCY

`TF(t, d) = count(t in d) / |d|`

×

INVERSE DOCUMENT FREQUENCY

`IDF(t) = log( N / df(t) )`

×

TF-IDF WEIGHT

`TF-IDF(t, d) = TF × IDF`

term	count in D1	TF = count / 3	df (of N = 4)	IDF = $\log_2$(N/df)	TF-IDF
dog	1	0.33	3	0.415	0.138
bites	1	0.33	2	1.000	**0.333**
man	1	0.33	3	0.415	0.138
eats · meat · food	0	0	2 · 1 · 1	—	0

bites is the most characteristic term of D1: just as frequent as dog and man inside the document, but present in half as many documents.

Notation and conventions

t = term · d = document · |d| = terms in d · N = documents in the corpus · df(t) = documents containing t. The base of the log only rescales the weights

Large TF-IDF = characteristic of this document. Near zero = absent, or so common it distinguishes nothing.

One matrix idea — different features, different values

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A DOCUMENT–TERM MATRIX (UNIGRAM VOCABULARY)

	dog	bites	man	eats	meat	food
D1	1	1	1	0	0	0
D2	1	1	1	0	0	0
D3	1	0	0	1	1	0
D4	0	0	1	1	0	1

WHAT CHANGES ACROSS REPRESENTATIONS

ONE-HOT
One token → one active dimension. Not yet one row per document.

BAG OF WORDS
Aggregate the token one-hots. Cells = counts, or 0/1 if binary.

BAG OF N-GRAMS
Change the columns: features become sequences such as "dog bites".

TF-IDF
Keep the same columns, replace counts with importance weights.

CHOICE 1 · FEATURES

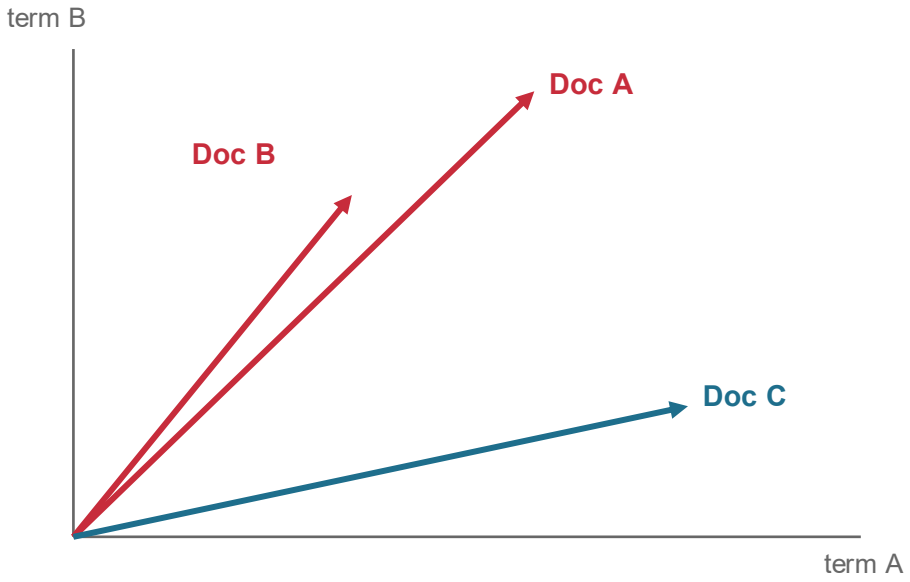
What does a column stand for?
unigrams · n-grams · characters

CHOICE 2 · WEIGHTS

What number goes in the cell?
binary · counts · TF-IDF

These two choices are independent — and together they define every representation we have seen today.

Cosine similarity: compare direction, not document length



Same direction = same feature proportions, whatever the length.

$$\cos(A, B) = (A \cdot B) / (\|A\| \|B\|)$$

1	same direction — same words in the same proportions
0	no shared features — orthogonal vectors
−1	opposite directions — impossible for counts or TF-IDF, which are never negative

For BoW and TF-IDF vectors the range is therefore 0 to 1.

Why not Euclidean distance?

Distance grows with document length: a long and a short document on the same topic sit far apart even when their word proportions match. Cosine divides that magnitude out. In the Wikipedia bias article, the first two sentences share so few words that their count vectors score only 0.117.

Cosine similarity is the default comparison for text vectors — today's counts, and the embeddings of Session 4.

A tiny search engine: vectorize the query, then rank

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QUERY: `"What is algorithmic bias?"` scored against 16 sentences from the Wikipedia article, unigram counts

S1	Algorithmic bias describes systematic and repeatable errors in a computer system…	`0.236`
S2	Bias can emerge due to many factors, including the design of the algorithm…	`0.125`
S3	Algorithmic bias has been cited in cases ranging from election outcomes…	`0.247`
S4	The study of algorithmic bias is most concerned with algorithms that reflect…	`0.433`

THE RANKING IS DEFENSIBLE — AND WRONG

S4 wins on shared words. But S1 is the actual definition, and it is what the question asked for. The vectorizer matched vocabulary, not meaning.

ONE FIX YOU ALREADY KNOW

With ngram_range=(1, 2), “algorithmic bias” becomes a single feature, and sentences that use the phrase rather than the two words separately move up the ranking.

Swap counts for TF-IDF, add bigrams, or change the tokenizer — retrieval quality is a representation problem before it is a model problem.

Quick check: which representation would you start with?

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A	Flag whether an email contains any of a short list of known policy terms.	***binary BoW?***
B	Distinguish “tax cuts increase growth” from “growth increases tax cuts”.	***bigrams?***
C	Rank documents for a query where common words should count for less.	***TF-IDF?***
D	Identify which of five languages a short message is written in.	***character n-grams?***

3 min in pairs → 4 min discussion. There is no universally best representation — only one that fits the task, the data and the budget.

Basic vectorization is powerful — and structurally limited

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01 DISCRETE UNITS

Words and n-grams are atomic, unrelated features. “run” and “ran” are no closer than “run” and “apple”.

02 SPARSE AND HIGH-DIMENSIONAL

Dimensionality grows with the vocabulary while nearly every cell stays zero — costly to store and harder to learn from.

03 ORDER IS PARTIAL OR LOST

BoW discards order entirely; n-grams recover only short local sequences, at a steep price in sparsity.

04 NO OOV HANDLING

A word absent from the fitted vocabulary has no dimension. The only remedy is to rebuild the vocabulary and refit.

NEXT SESSION → DISTRIBUTED REPRESENTATIONS

Instead of one dimension per vocabulary item, learn dense low-dimensional vectors in which related words sit close together — so that “run” and “ran” are near each other, and an unseen word is no longer a dead end.

TF-IDF is still a strong first solution for many tasks — and still the baseline a neural model has to beat.

Keep this mental model

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TOKENS	→	VOCABULARY	→	WEIGHTS	→	VECTOR	→	USE
what are the units?		what gets a dimension?		presence · counts · TF-IDF		one row per document		similarity · classification

1. Representation is a modeling decision: the vector space you choose sets the ceiling for every model built on it.

2. Counts answer “how often here”; TF-IDF adds “how rare elsewhere”. Features and weights are two separate choices.

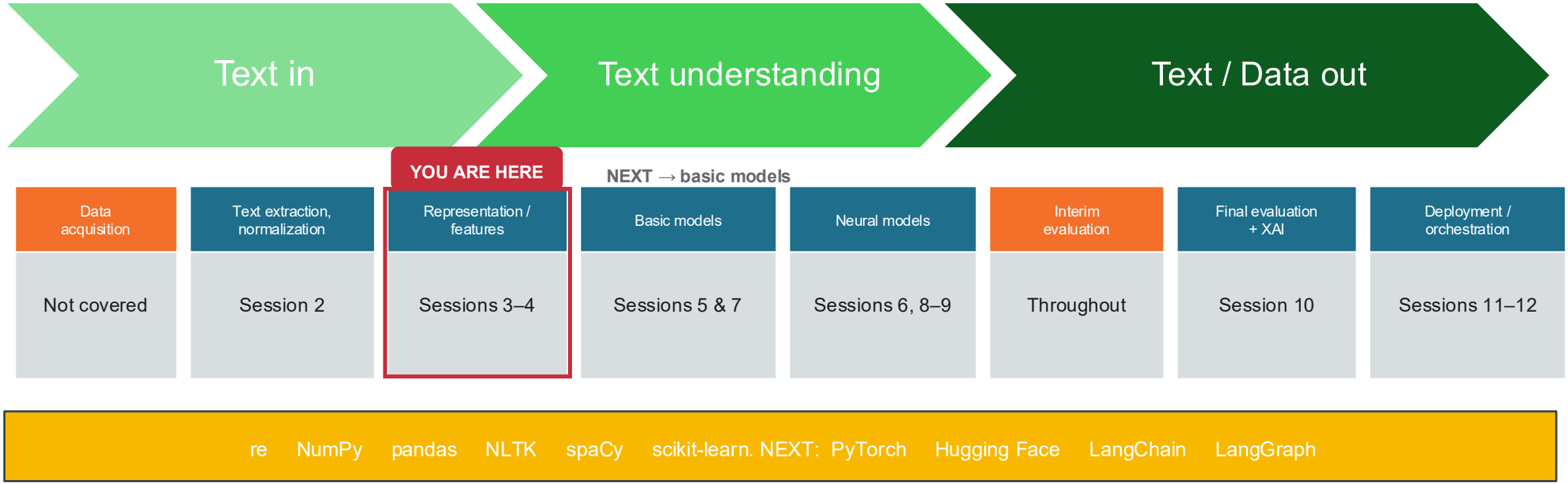
3. Cosine similarity compares direction, not length — which is what makes these vectors usable for retrieval.

The recurring question

What does this representation make visible to the model — and what does it hide?

Next session → distributed representations: dense vectors where similar words can sit close together.

Where we are in the NLP pipeline



Today: tokens → vocabulary → counts → weights → vectors. Next: learn the dimensions instead of assigning them.

THANK YOU!

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one-hot → bag of words → n-grams → TF-IDF

Every one of them answers the same question:
which dimension, and what number?

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